Code Coverage and Branch Coverage Report

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Record	Verified value
Project	Collaborative Study Material & Quiz Platform (Study Commons)
Prepared by	Harshit Chandak - PES2UG24CS185; whole Team 10 evidence
Execution date	2026-11-05 (Asia/Calcutta)
PDF preparation date	2026-11-06; packaging existing results, not a new test run
Tested source	888c0666ae78e14be134a6c090be860d254ddb00
Integrated main commit	1c6ecafc3807a1f3916d79ea81416cd580e12eec (PR #44, merged and closed)
Preserved baseline	Submitted SRS, SAD and Test Plan unchanged

Scope covers Dharani (Auth/Materials/Dashboard), Prajwal (Question Bank/Quiz), Karan (Attempt/Result/History/Performance), and Harshit (Admin/Security/CI/shared integration).

Evidence links in this PDF point to immutable files in the integrated repository. Dates, environment, coverage limitations and outstanding acceptance items are retained; no approval is inferred from automated passes.

One combined PDF for the third form field. Part 1 reports code coverage (lines/statements/functions and summary branches); Part 2 reports branch outcomes and per-file gaps. Both use the same 5 November V8 measurements. Weighted totals sum covered and measured counters across backend/frontend rather than averaging percentages.

Part 1 - Code Coverage Report

Execution date: **2026-11-05**, Asia/Calcutta. Source: `888c0666ae78e14be134a6c090be860d254ddb00`. Branch: `PES2UG24CS185-Sprint-2`. The execution manifest identifies the tested source and reporting configuration.

Scope: **the whole Team 10 project**, covering Auth/Materials (Dharani), Question Bank/Quiz (Prajwal), Attempt/Result/History/Performance (Karan), Admin/Security/CI and shared integration (Harshit). Execution coordination does not replace independent peer acceptance.

Measured by Vitest 5.0.3 and its version-matched V8 provider during the backend/frontend suites. Browser coverage is not merged into these numbers.

Layer	Metric	Covered	Total	Percent
backend	lines	593	606	97.85
backend	statements	672	689	97.53
backend	functions	183	185	98.91
backend	branches	427	464	92.02
frontend	lines	1014	1241	81.7
frontend	statements	1234	1569	78.64
frontend	functions	329	415	79.27
frontend	branches	998	1386	72

Backend scope: `backend/src/**/*.ts`, excluding the server entry point and type-only directory, as configured in `vitest.config.ts`. Backend gates: lines/statements/functions >=85%, branches >=80%; all passed. Frontend scope: all `frontend/src/**/*.{ts,tsx}`, excluding test files, test setup and declarations; unused/legacy UI and the application entry point remain included. No frontend numeric gate was previously defined; these measured results establish the baseline, not an implied backend-threshold pass. CSS/assets and deployment scripts are outside V8 source coverage.

Evidence: sanitized per-file backend summary, frontend summary, backend LCOV, frontend LCOV. LCOV records identify unexecuted lines and branch counters. Lower frontend service percentages can coexist with passing browser journeys because browser execution is not included here.

Layer	Source	Lines %	Statements %	Functions %	Branches %
backend	backend/src/admin/moderation.ts	85.18	87.5	85.71	83.33
backend	backend/src/app.ts	100	100	100	100
backend	backend/src/attempts/evaluate.ts	100	100	100	100
backend	backend/src/attempts/history-route.ts	91.66	91.66	100	83.33
backend	backend/src/attempts/history.ts	0	0	0	0
backend	backend/src/attempts/mongo-repository.ts	95.23	93.47	100	75.75
backend	backend/src/attempts/router.ts	100	100	100	100
backend	backend/src/attempts/submission-service.ts	96.15	96.15	100	93.75
backend	backend/src/attempts/trusted-evaluation.ts	96	96	100	91.66
backend	backend/src/attempts/validation.ts	100	100	100	100
backend	backend/src/config.ts	100	100	100	100
backend	backend/src/db.ts	60	60	100	100
backend	backend/src/errors.ts	100	100	100	95.65
backend	backend/src/materials/upload.ts	100	98.79	95.23	94
backend	backend/src/middleware/auth.ts	100	100	100	100
backend	backend/src/middleware/validation.ts	100	100	100	100
backend	backend/src/models/question.ts	100	100	100	100
backend	backend/src/models/quiz-attempt.ts	100	100	100	100
backend	backend/src/models/quiz.ts	100	100	100	100
backend	backend/src/models/user.ts	100	100	100	100
backend	backend/src/questions/filters.ts	100	100	100	100
backend	backend/src/questions/router.ts	100	100	100	80
backend	backend/src/questions/serializer.ts	100	100	100	100
backend	backend/src/questions/validation.ts	100	100	100	87.5
backend	backend/src/quizzes/filters.ts	100	100	100	100
backend	backend/src/quizzes/router.ts	97.77	97.95	100	71.42
backend	backend/src/quizzes/validation.ts	100	100	100	83.33
backend	backend/src/repositories/mongo-account-repository.ts	100	100	100	100
backend	backend/src/routes/auth.ts	100	100	100	100

Layer	Source	Lines %	Statements %	Functions %	Branches %
backend	backend/src/runtime.ts	100	100	100	100
backend	backend/src/security/auth-service.ts	100	93.33	100	92.59
backend	backend/src/security/password.ts	100	100	100	100
backend	backend/src/security/provision-admin.ts	96	96.87	100	87.09
backend	backend/src/security/sessions.ts	100	100	100	100
frontend	frontend/src/App.tsx	71.05	72.5	64.7	75.86
frontend	frontend/src/components/FormField.tsx	100	100	100	100
frontend	frontend/src/components/MaterialLibrary.tsx	0	0	0	0
frontend	frontend/src/components/Navbar.tsx	92.3	94.44	100	88.46
frontend	frontend/src/components/ThemeToggle.tsx	100	88.23	87.5	80
frontend	frontend/src/hooks/useMaterialLibrary.ts	100	96.29	100	77.27
frontend	frontend/src/main.tsx	0	0	100	100
frontend	frontend/src/pages/AdminDashboardPage.tsx	100	100	100	80
frontend	frontend/src/pages/AdminPage.tsx	86.59	86.23	86.36	77.23
frontend	frontend/src/pages/CreateQuizPage.tsx	92.3	92.68	85.18	90.9
frontend	frontend/src/pages/HomePage.tsx	100	100	100	100
frontend	frontend/src/pages/LoginPage.tsx	100	100	100	83.33
frontend	frontend/src/pages/MaterialsPage.tsx	82.67	75.11	70	66.12
frontend	frontend/src/pages/QuestionBankPage.tsx	80.73	80.86	78.37	78.57
frontend	frontend/src/pages/QuizAttemptLoader.tsx	95.45	88	88.88	75
frontend	frontend/src/pages/QuizAttemptPage.tsx	77.66	77.58	72.41	74.75
frontend	frontend/src/pages/QuizHistoryPage.tsx	96.66	93.75	92.85	67.85
frontend	frontend/src/pages/QuizInsightsPage.tsx	96	96.66	93.33	68.18
frontend	frontend/src/pages/QuizListPage.tsx	89.13	89.13	91.66	78.72
frontend	frontend/src/pages/QuizResultPage.tsx	94.87	92.68	100	93.54
frontend	frontend/src/pages/QuizViewPage.tsx	100	88	100	69.44
frontend	frontend/src/pages/RegisterPage.tsx	100	100	100	90
frontend	frontend/src/pages/StudentDashboardPage.tsx	100	100	100	62.5
frontend	frontend/src/services/attemptHistoryCsv.ts	41.17	47.61	83.33	50
frontend	frontend/src/services/attemptHistoryService.ts	90	84.44	87.5	90.76
frontend	frontend/src/services/attemptInsights.ts	100	100	100	75
frontend	frontend/src/services/attemptService.ts	91.3	84	50	71.05
frontend	frontend/src/services/authService.ts	78.94	75.6	67.74	64.4
frontend	frontend/src/services/authValidation.ts	100	80	100	84.37
frontend	frontend/src/services/questionService.ts	50	40	25	33.33
frontend	frontend/src/services/quizAttemptDataService.ts	80	80	87.5	75.71

Layer	Source	Lines %	Statements %	Functions %	Branches %
frontend	frontend/src/services/quizDraft.ts	100	100	100	100
frontend	frontend/src/services/quizService.ts	11.62	9.09	12.5	6.66
frontend	frontend/src/services/recentMaterials.ts	91.66	77.77	100	82.35

Whole measured project totals

Counters below sum the backend and frontend measurements; percentages are covered / total, rounded to two decimals. They use the declared instrumentation scope and do not include browser execution.

Metric	Covered	Total	Percent
lines	1607	1847	87.01
statements	1906	2258	84.41
functions	512	600	85.33
branches	1425	1850	77.03

Part 2 - Branch Coverage Report

Execution date: **2026-11-05**, Asia/Calcutta. Source: `888c0666ae78e14be134a6c090be860d254ddb00`. Branch: `PES2UG24CS185-Sprint-2`. The execution manifest identifies the tested source and reporting configuration.

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Branch coverage measures executed decision outcomes; it is not Git branch count, requirement coverage or MC/DC. V8 counters and LCOV BRDA records are retained separately for both layers.

Layer	Covered	Total	Uncovered	Percent
backend	427	464	37	92.02
frontend	998	1386	388	72

The backend 80% branch gate passed. No frontend gate is claimed. Remaining unexecuted outcomes are coverage gaps, not automatically confirmed defects; prioritize shared authentication, moderation and Quiz service error paths for further tests. Do not remove uncovered product code from measurement just to increase percentages.

Layer	Source	Covered	Total	Uncovered	Percent
backend	backend/src/admin/moderation.ts	10	12	2	83.33
backend	backend/src/app.ts	30	30	0	100
backend	backend/src/attempts/evaluate.ts	8	8	0	100
backend	backend/src/attempts/history-route.ts	5	6	1	83.33
backend	backend/src/attempts/history.ts	0	0	0	0
backend	backend/src/attempts/mongo-repository.ts	25	33	8	75.75
backend	backend/src/attempts/router.ts	6	6	0	100
backend	backend/src/attempts/submission-service.ts	15	16	1	93.75
backend	backend/src/attempts/trusted-evaluation.ts	11	12	1	91.66
backend	backend/src/attempts/validation.ts	2	2	0	100
backend	backend/src/config.ts	11	11	0	100
backend	backend/src/db.ts	0	0	0	100
backend	backend/src/errors.ts	22	23	1	95.65
backend	backend/src/materials/upload.ts	47	50	3	94
backend	backend/src/middleware/auth.ts	12	12	0	100
backend	backend/src/middleware/validation.ts	2	2	0	100
backend	backend/src/models/question.ts	13	13	0	100
backend	backend/src/models/quiz-attempt.ts	4	4	0	100
backend	backend/src/models/quiz.ts	20	20	0	100
backend	backend/src/models/user.ts	2	2	0	100
backend	backend/src/questions/filters.ts	13	13	0	100
backend	backend/src/questions/router.ts	16	20	4	80
backend	backend/src/questions/serializer.ts	19	19	0	100

Layer	Source	Covered	Total	Uncovered	Percent
backend	backend/src/questions/validation.ts	7	8	1	87.5
backend	backend/src/quizzes/filters.ts	15	15	0	100
backend	backend/src/quizzes/router.ts	20	28	8	71.42
backend	backend/src/quizzes/validation.ts	5	6	1	83.33
backend	backend/src/repositories/mongo-account-repository.ts	6	6	0	100
backend	backend/src/routes/auth.ts	1	1	0	100
backend	backend/src/runtime.ts	9	9	0	100
backend	backend/src/security/auth-service.ts	25	27	2	92.59
backend	backend/src/security/password.ts	17	17	0	100
backend	backend/src/security/provision-admin.ts	27	31	4	87.09
backend	backend/src/security/sessions.ts	2	2	0	100
frontend	frontend/src/App.tsx	22	29	7	75.86
frontend	frontend/src/components/FormField.tsx	8	8	0	100
frontend	frontend/src/components/MaterialLibrary.tsx	0	23	23	0
frontend	frontend/src/components/Navbar.tsx	23	26	3	88.46
frontend	frontend/src/components/ThemeToggle.tsx	8	10	2	80
frontend	frontend/src/hooks/useMaterialLibrary.ts	17	22	5	77.27
frontend	frontend/src/main.tsx	0	0	0	100
frontend	frontend/src/pages/AdminDashboardPage.tsx	20	25	5	80
frontend	frontend/src/pages/AdminPage.tsx	95	123	28	77.23
frontend	frontend/src/pages/CreateQuizPage.tsx	60	66	6	90.9
frontend	frontend/src/pages/HomePage.tsx	0	0	0	100
frontend	frontend/src/pages/LoginPage.tsx	10	12	2	83.33
frontend	frontend/src/pages/MaterialsPage.tsx	121	183	62	66.12
frontend	frontend/src/pages/QuestionBankPage.tsx	66	84	18	78.57
frontend	frontend/src/pages/QuizAttemptLoader.tsx	9	12	3	75
frontend	frontend/src/pages/QuizAttemptPage.tsx	77	103	26	74.75
frontend	frontend/src/pages/QuizHistoryPage.tsx	19	28	9	67.85
frontend	frontend/src/pages/QuizInsightsPage.tsx	15	22	7	68.18
frontend	frontend/src/pages/QuizListPage.tsx	37	47	10	78.72
frontend	frontend/src/pages/QuizResultPage.tsx	58	62	4	93.54
frontend	frontend/src/pages/QuizViewPage.tsx	25	36	11	69.44
frontend	frontend/src/pages/RegisterPage.tsx	9	10	1	90
frontend	frontend/src/pages/StudentDashboardPage.tsx	5	8	3	62.5
frontend	frontend/src/services/attemptHistoryCsv.ts	6	12	6	50
frontend	frontend/src/services/attemptHistoryService.ts	59	65	6	90.76
frontend	frontend/src/services/attemptInsights.ts	9	12	3	75

Layer	Source	Covered	Total	Uncovered	Percent
frontend	frontend/src/services/attemptService.ts	27	38	11	71.05
frontend	frontend/src/services/authService.ts	76	118	42	64.4
frontend	frontend/src/services/authValidation.ts	27	32	5	84.37
frontend	frontend/src/services/questionService.ts	9	27	18	33.33
frontend	frontend/src/services/quizAttemptDataService.ts	53	70	17	75.71
frontend	frontend/src/services/quizDraft.ts	11	11	0	100
frontend	frontend/src/services/quizService.ts	3	45	42	6.66
frontend	frontend/src/services/recentMaterials.ts	14	17	3	82.35

Detailed line/branch counters: backend LCOV, frontend LCOV.

Whole measured project: **1425/1850 branches (77.03%)**. This weighted total uses both layers; the existing 80% CI branch threshold applies to the backend only.